

SUMMARY

AI Engineer with production-grade experience building end-to-end generative AI systems, from design through deployment. Proficient in LangChain, Vertex AI (Gemini), RAG architecture, LLM evaluation, and agentic workflows. Skilled in designing scalable APIs, implementing safety validation frameworks, monitoring production systems, and optimizing performance at scale. Seeking AI Engineer roles to build reliable, production-ready AI-powered products with strong software engineering practices.

EDUCATION

University of South Florida, Tampa, FL

Aug 2024 – May 2026

Master of Science, Artificial Intelligence and Business Analytics

GPA: 3.95 / 4.0

Coursework: Machine Learning, Business Statistics, Advanced Database Management Systems, Data Visualization, Big Data Analytics, Data Warehousing, Application Development for Analytics

Malla Reddy Engineering College for Women, Hyderabad, India

Jun 2020 – Apr 2024

Bachelor of Technology, Computer Science Engineering - Data Science

GPA: 3.7 / 4.0

TECHNICAL SKILLS

Generative AI & LLMs: LangChain (Chains, Agents, Tools, LCEL), Vertex AI (Gemini), OpenAI API, Prompt Engineering, RAG/Vector Search, Embeddings, LLM Evaluation Metrics (BLEU, ROUGE, Custom Evaluation Frameworks), Hallucination Detection & Mitigation, Hugging Face Transformers, Model Fine-tuning

AI/ML Engineering: Scikit-learn, PyTorch, TensorFlow, Classification, Regression, Ensemble Methods, Feature Engineering, Hyperparameter Tuning, NLP/Text Preprocessing, Named Entity Recognition, Cross-Validation, Model Evaluation

Software Engineering & APIs: FastAPI, Flask, RESTful API Design, Request Validation (Pydantic), Error Handling, Rate Limiting, Python (Pandas, NumPy), SQL, Git/GitHub, Code Quality (linting, type hints)

Production & MLOps: Docker & Docker Compose, Model Serving, API Monitoring & Logging (Cloud Logging), Data Validation Pipelines, SQL Safety Checks, Performance Optimization, Containerization Best Practices, Structured Outputs

Cloud Platforms: Google Cloud Platform (Vertex AI, Cloud Run, Cloud Logging, BigQuery), Infrastructure as Code Concepts

BI & Visualization: Tableau, Power BI (DAX), Matplotlib, Seaborn, Dashboard Design

EXPERIENCE

Project Assistant - University of South Florida

Feb 2025 – Sep 2025

- Designed and deployed a Python-based ETL pipeline standardizing 180+ country name variations (EU, USSR, UAE) into ISO codes using fuzzy matching (RapidFuzz) and custom override dictionaries, reducing data mismatch errors by ~90%
- Automated end-to-end data processing workflows ingestion, matching, standardization, validation, and mismatch flagging cutting manual data preparation time by an estimated 70%
- Integrated U.S. Census Bureau ACS 5-Year API using FIPS-based conditional logic to extract demographic and income data at ZIP, county, and state granularity, producing warehouse-ready outputs consumed by Tableau BI dashboards
- Engineered structured datasets aligned with data warehouse standards and schema conventions, supporting downstream reporting and visualization workflows for 3+ stakeholder teams

Data Science Intern - Accenture

Jun 2023 – May 2024

- Supported ML initiatives through data preprocessing, feature engineering, and EDA on cognitive and socioeconomic datasets (5M+ records); identified and resolved data quality issues affecting model performance
- Developed Tableau and Power BI dashboards translating ML model outputs into actionable KPIs for C-level stakeholders; automated reporting reducing manual reporting time by 30%
- Built SQL-based reporting pipelines analyzing operational trends and improving feature consistency across ML workflows; designed and documented data models for 8+ downstream analytics use cases

Research Assistant - MRECW

Jun 2022 – Jun 2023

- Designed and deployed an end-to-end NLP pipeline using Python, Scikit-learn, and NLTK for preprocessing and feature extraction from 500+ unstructured construction incident reports
- Trained and evaluated ensemble ML classifiers (Random Forest, Gradient Boosting, SVM) to identify high-risk workplace scenarios with 92% accuracy, enabling data-driven safety analysis and predictive risk assessment
- Generated actionable safety insights that contributed to a measurable 20% reduction in reported workplace accidents; published research findings in peer-reviewed venue

PROJECTS

AI-Powered SQL Analytics Assistant | LangChain, Vertex AI (Gemini), RAG, Python, SQL

- Built production-ready NL-to-SQL engine with schema-aware prompting, leveraging live database metadata to convert natural language questions into context-aware, validated SQL queries
- Engineered automated SQL validation guardrails (SELECT-only enforcement, safe join patterns, LIMIT compliance) with modular validation layer preventing unsafe query execution
- Developed Streamlit-based self-service analytics UI with interactive query results, automated summaries, and chart visualizations for non-technical end-users
- Architected modular LLM pipeline with Vertex AI Gemini integration, local fallback support, and cost-aware AI design reflecting real-world production constraints

RAG-based Document Q&A API Service | LangChain, FastAPI, PostgreSQL (pgvector), Vertex AI, Pytest

- Built production-ready RAG service: document ingestion (PDF/text parsing), embedding generation (Vertex AI), vector storage (PostgreSQL pgvector), retrieval-augmented generation
- Designed FastAPI endpoints with Pydantic validation, error handling, rate limiting, and Cloud Logging for production monitoring; achieved <100ms response latency
- Implemented LLM evaluation framework measuring answer relevance (ROUGE), fact consistency, retrieval quality; pytest unit/integration tests with performance benchmarks

CERTIFICATIONS

Data Science Fundamentals with Python and SQL – IBM | Introduction to Data Science – IBM | Microsoft Power BI Data Analyst (PL-300) | Programming Essentials in Python – Cisco (OpenEDG) | Python (Basic) – HackerRank